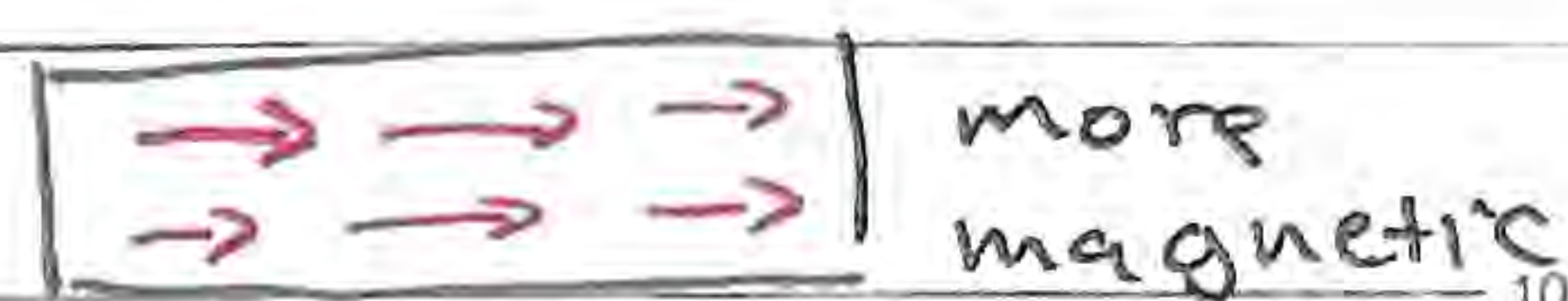


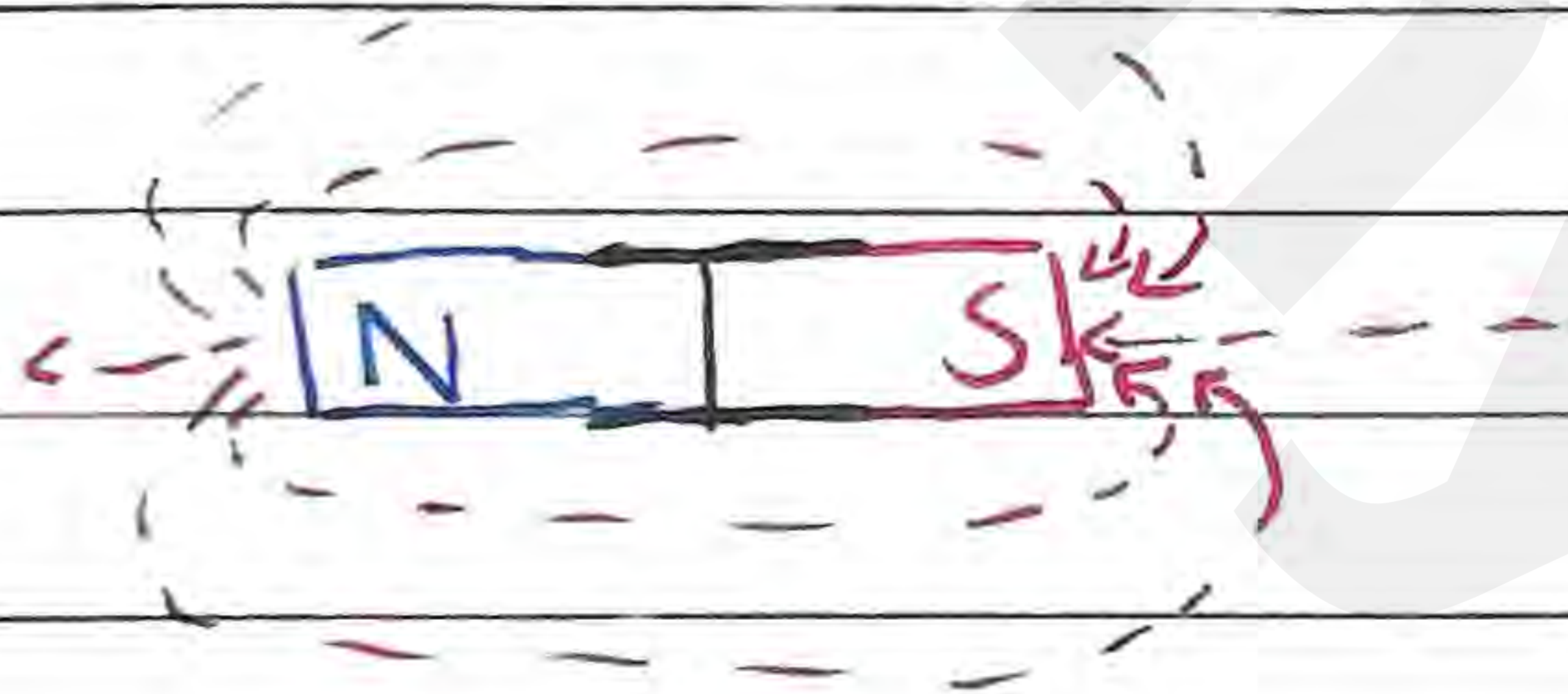
Magnetism

Every object has some form of magnetism, though to varying degrees

- This results from electron spins in magnetic domains. An object with electron spins aligned easily is called ferromagnetic



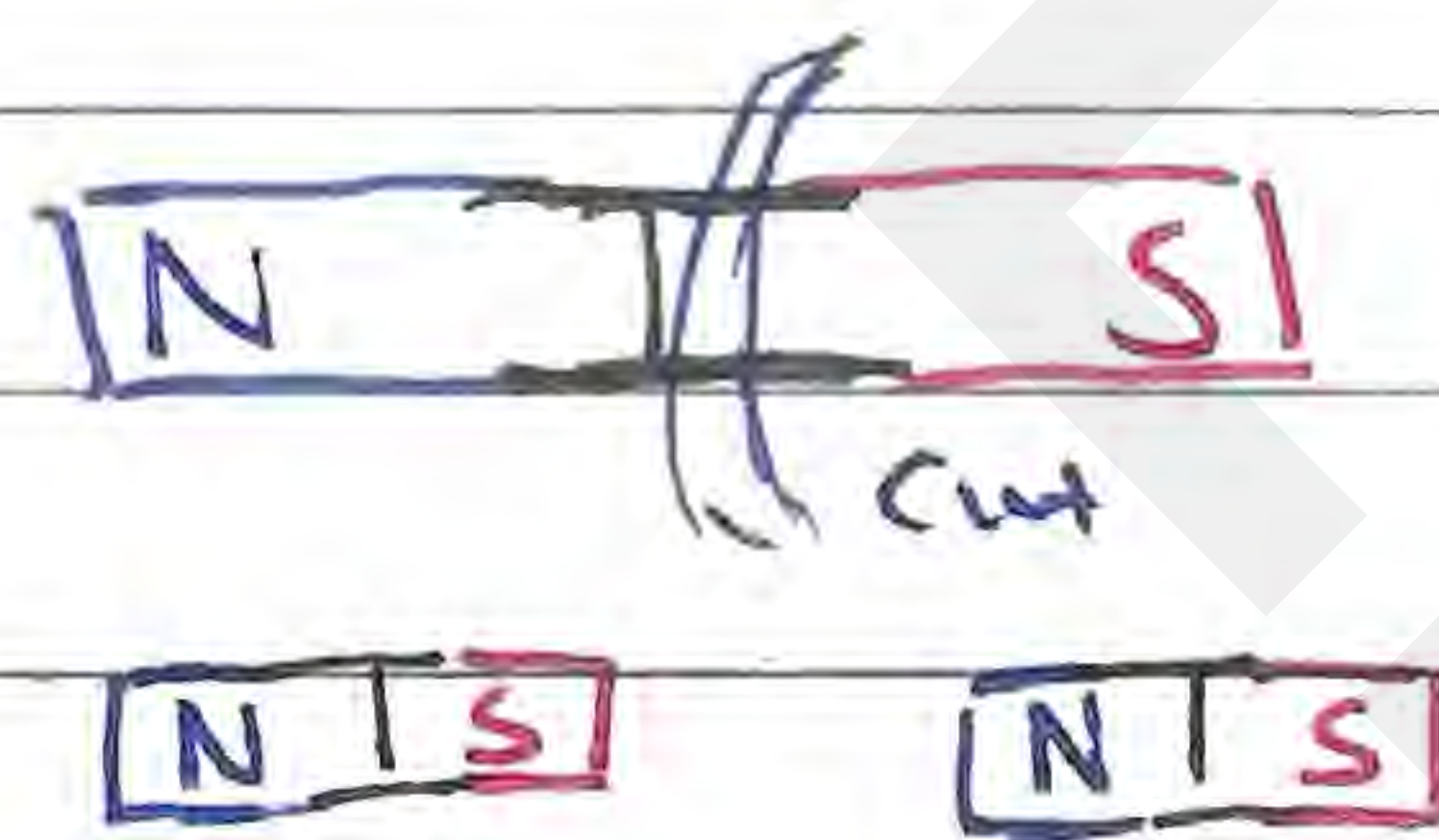
The two directions are North and South



They are not the same property as charge

A magnet, when split, will also have two poles

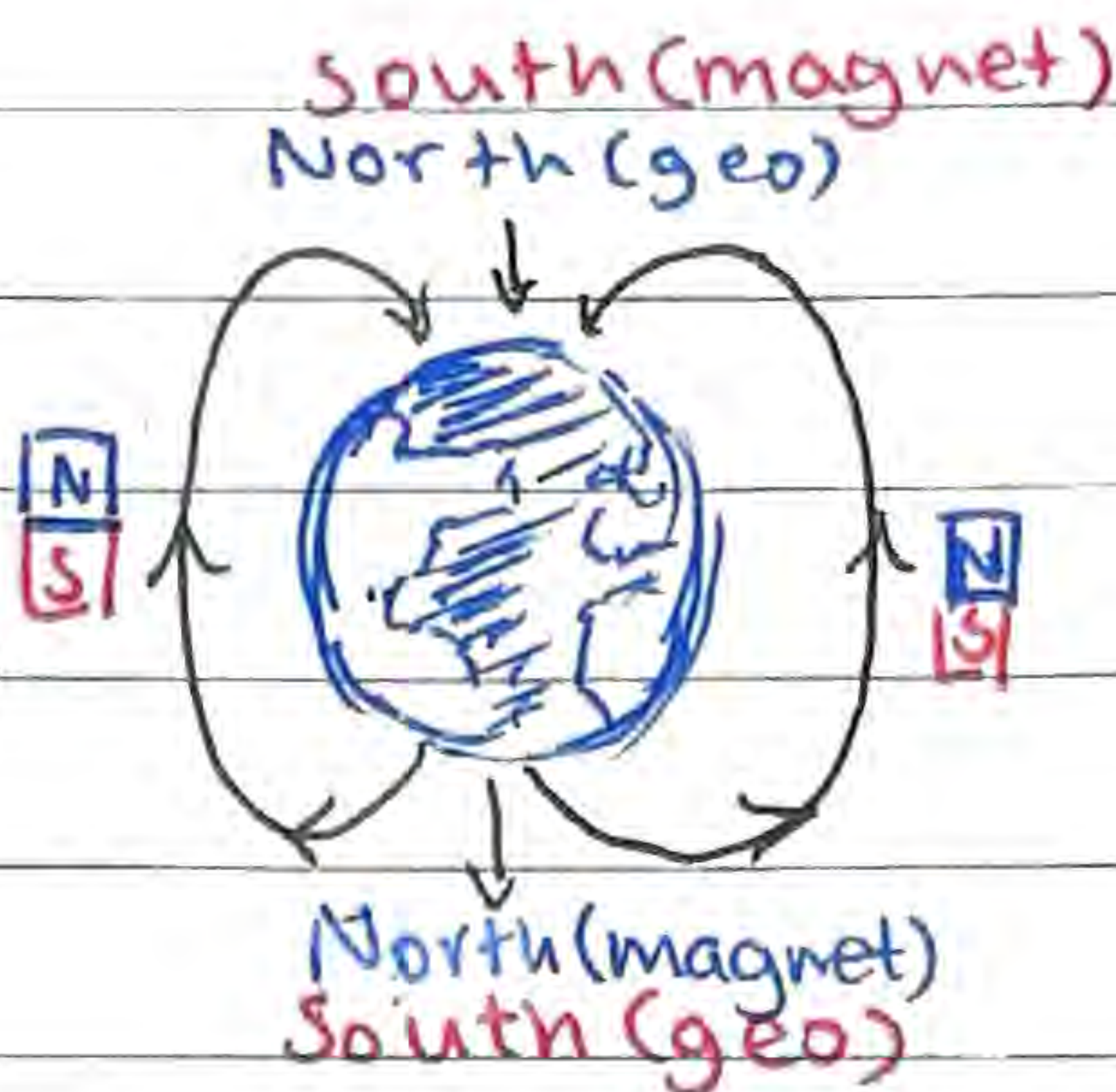
Can be cut infinitely
but still will have two poles in
each piece



A compass works by aligning itself with Earth's magnetic field

Magnetism

The north of the compass aligns to Earth's geographic north pole

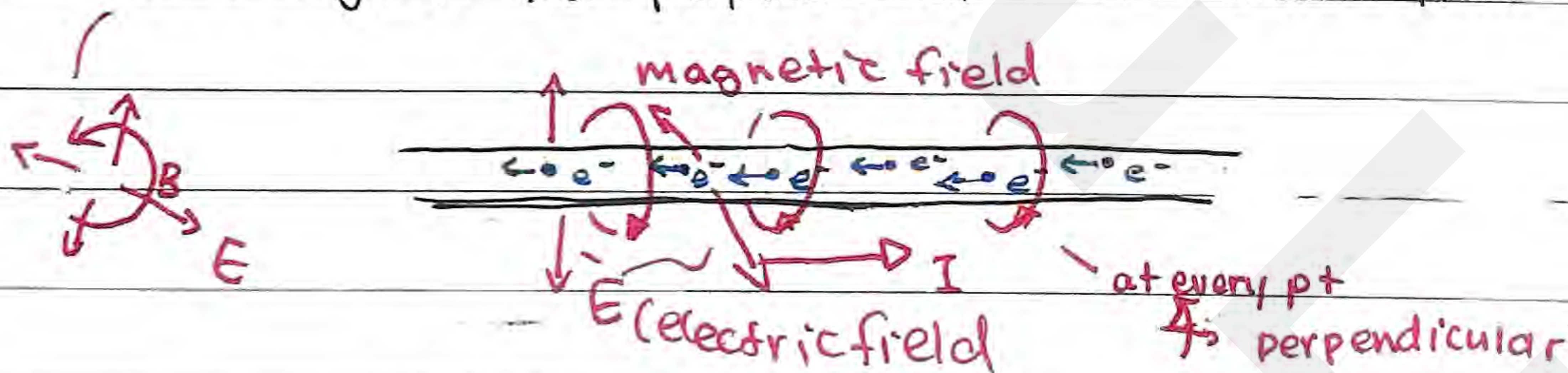


But that indicates that Earth's geographic North is actually its magnetic south!

Magnetic fields

Know that magnetic fields are perpendicular to the electric field (recall electromagnetic waves!)

Thus, in a wire such that an electric field exists, there will be a magnetic field perpendicular to the electric field



The specific direction of the magnetic field can be found with the right-hand rule (cross product)



B represents magnetic field

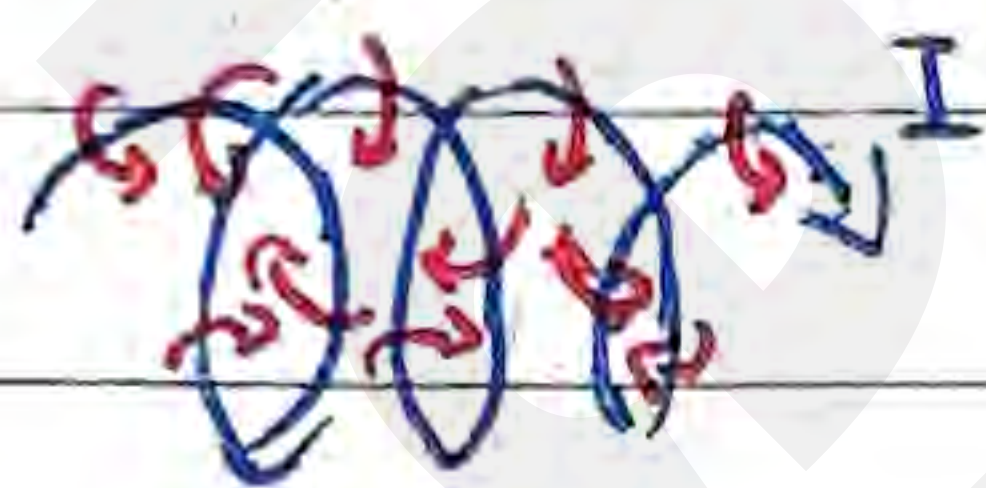
$B \propto \frac{1}{r}$, further away from wire, weaker magnetic field

Electromagnet

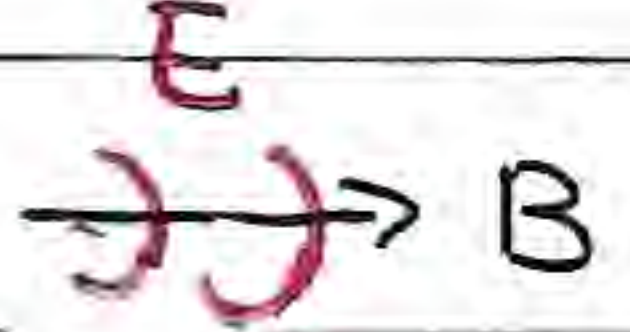
In a single wire, electric field is straight, magnetic field is orbiting wire.

How can we reverse this?

we can make the electric field orbit a straight, imaginary line . . .



in the center, all magnetic fields point in the same direction!

So in an electromagnet (like a coil)  $E \rightarrow B$

- In this case, North represents direction that the magnetic field points to, South is the direction the field points from.



The direction of the magnetic field is not determined by the net direction of current, rather the counterclockwise or clockwise rotation

↳ Use right-hand rule



Magnetic Field

Representation of 3D vectorial quantities in a 2D plane



We use arrows to represent planar vectors

But when they cross our viewing plane, we use dots and crosses.

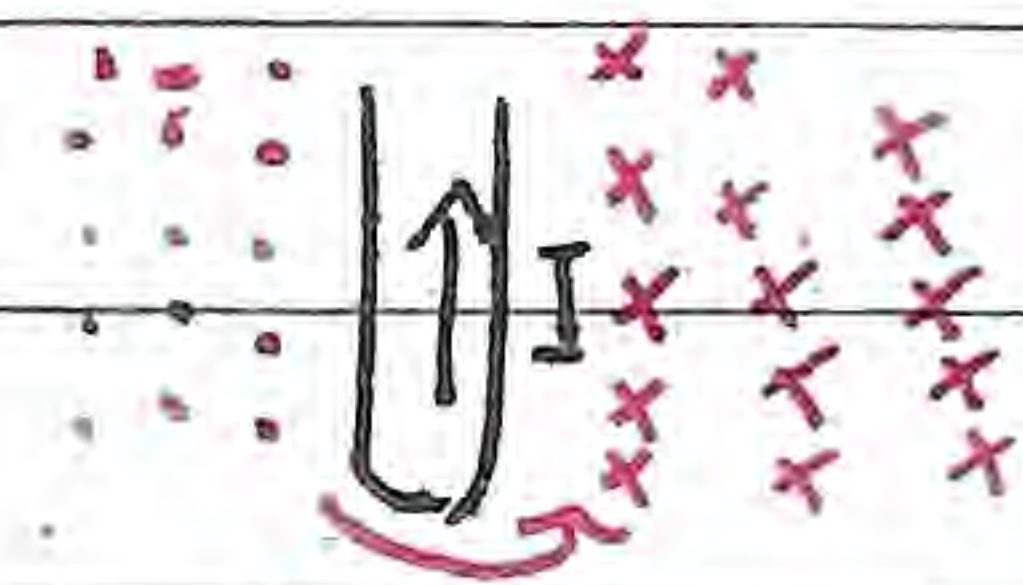
out of



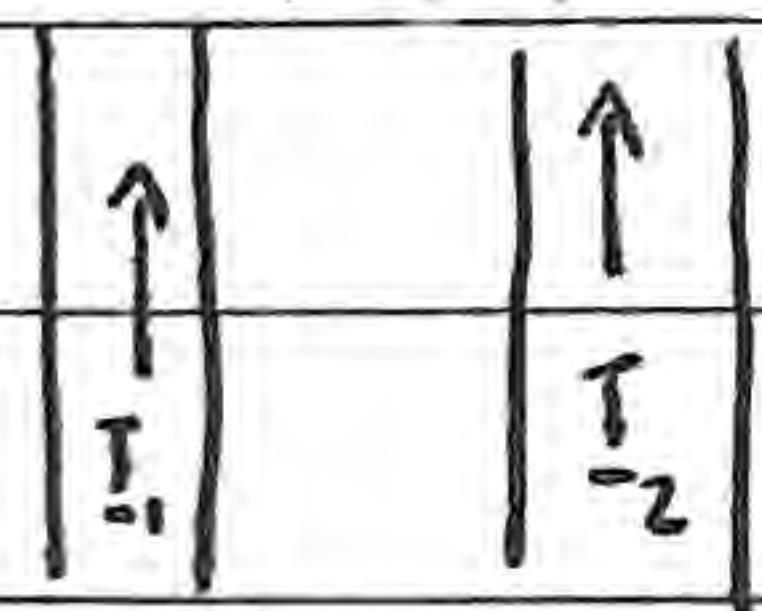
into



This is used to describe electric fields, among other vectorial quantities



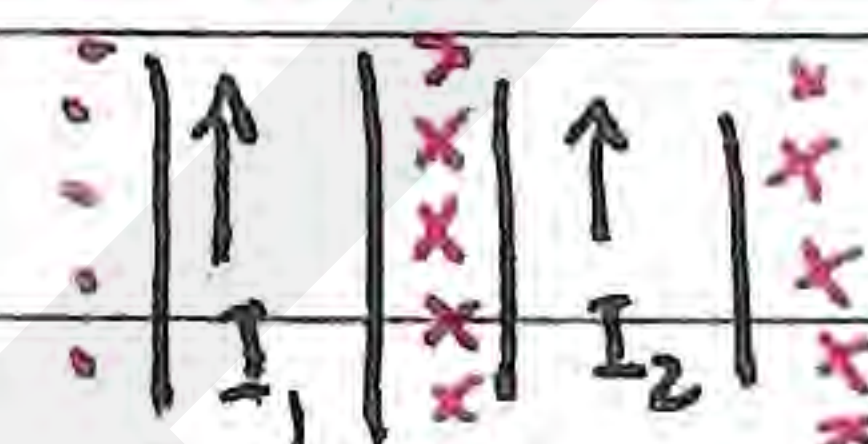
Magnetic Force between parallel wires



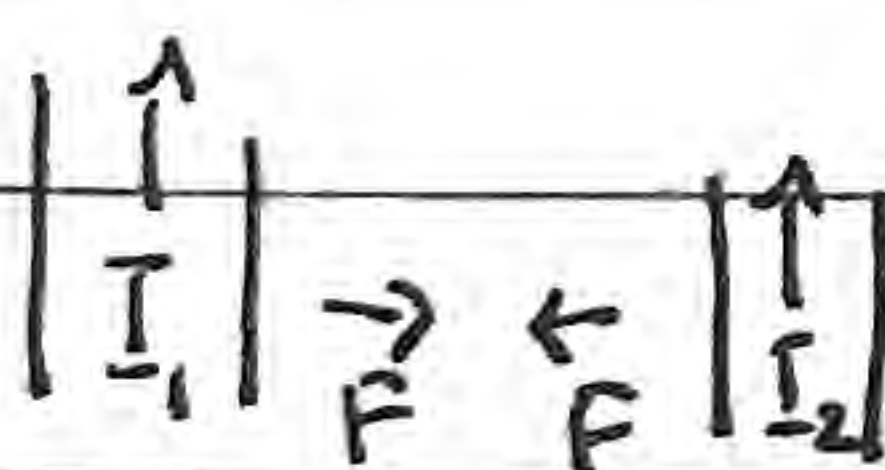
repulsive or

attractive? → we break this down into two

field of I_1



Left hand rule, Force by I_1 on I_2

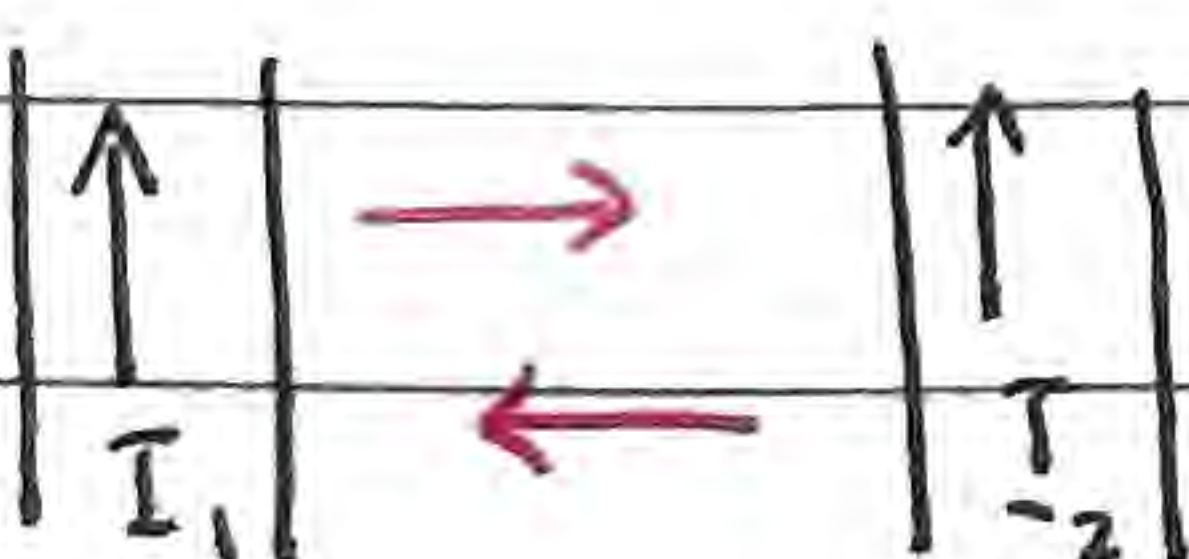


Thus



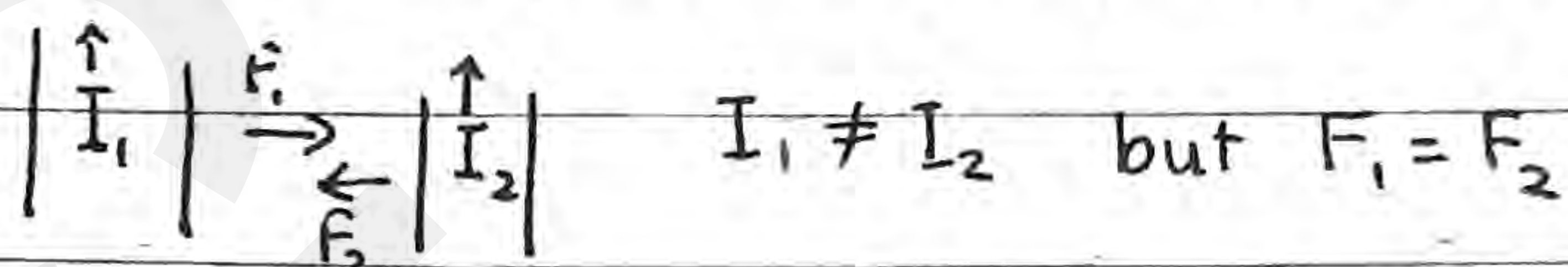
$F \rightarrow$ Force by I_2 on I_1

As such, when I in same direction, they attract. Else, they repulse



Magnetic Force on parallel wires

According to Newton's Third Law, attractive/repulsive forces are equal (even if current is different)



$$| \vec{I}_1 | \xrightarrow{F_1} | \vec{I}_2 | \quad I_1 \neq I_2 \text{ but } F_1 = F_2$$

But how much is the force specifically?

$$\frac{F}{L} = \frac{\mu_0 \cdot I_1 \cdot I_2}{2\pi \cdot r}$$

where ...

μ_0 - Permeability of free space ($4\pi \times 10^{-7} \text{ N} \cdot \text{A}^{-2}$)

I_i - Current of wire i

L - Length of wire

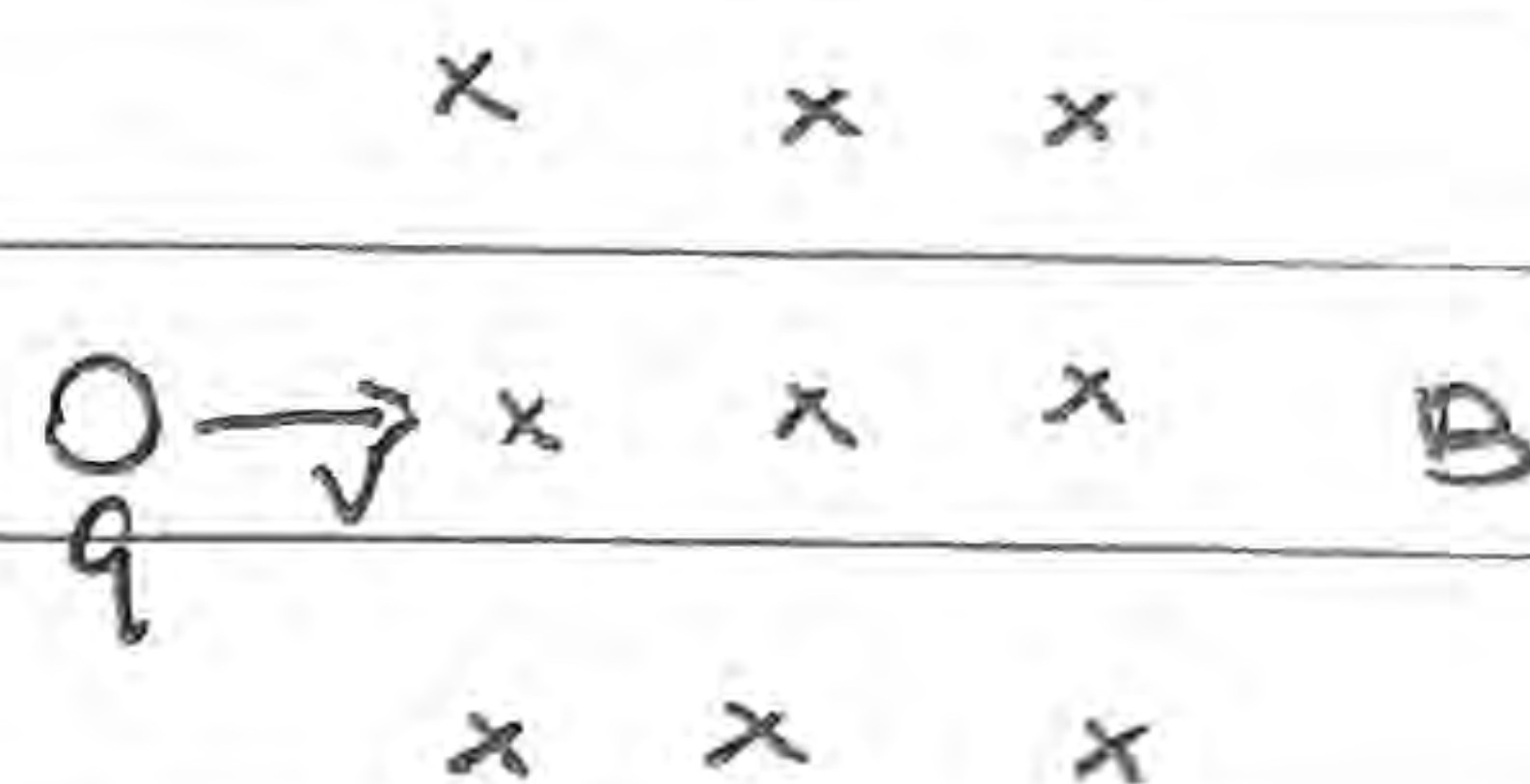
F - Magnitude of resulting magnetic force

r - Distance between wires

When you have more than two parallel wires, you must add the magnetic forces together or subtract depending on attraction or repulsion

Magnetic Force on a moving charge

when a charged particle enters a magnetic field, magnetic force is done on the charge



$$F = q \cdot v \cdot B \cdot \sin \theta$$

where...

F - Magnitude of force done on charge

q - Charge of moving particle

v - Velocity of moving charge

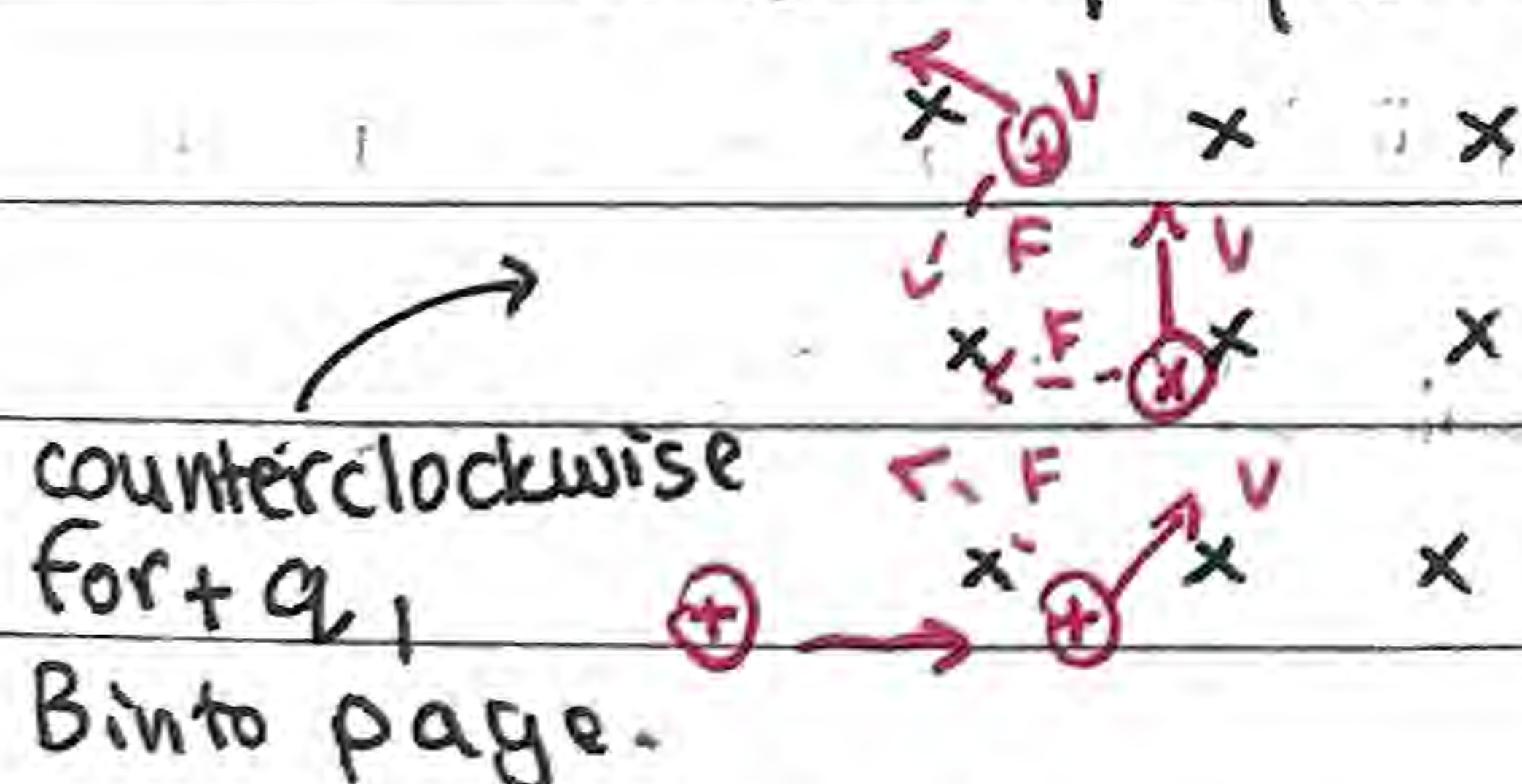
B - Magnetic field magnitude

θ - Angle between v and B

The direction of the force can be found with the left-hand rule

Perfectly circular motion

Since force direction is not the same velocity, it is a perpetual process that changes its trajectory



counterclockwise for +q, B into page.

Clockwise for +q, B out of page

As such, it moves in circular motion where magnetic force is centripetal

Recalling $F = m \cdot a$ and centripetal acceleration is $a = \frac{v^2}{r}$, we get... $r = \frac{m v}{q B}$

radius of circular motion

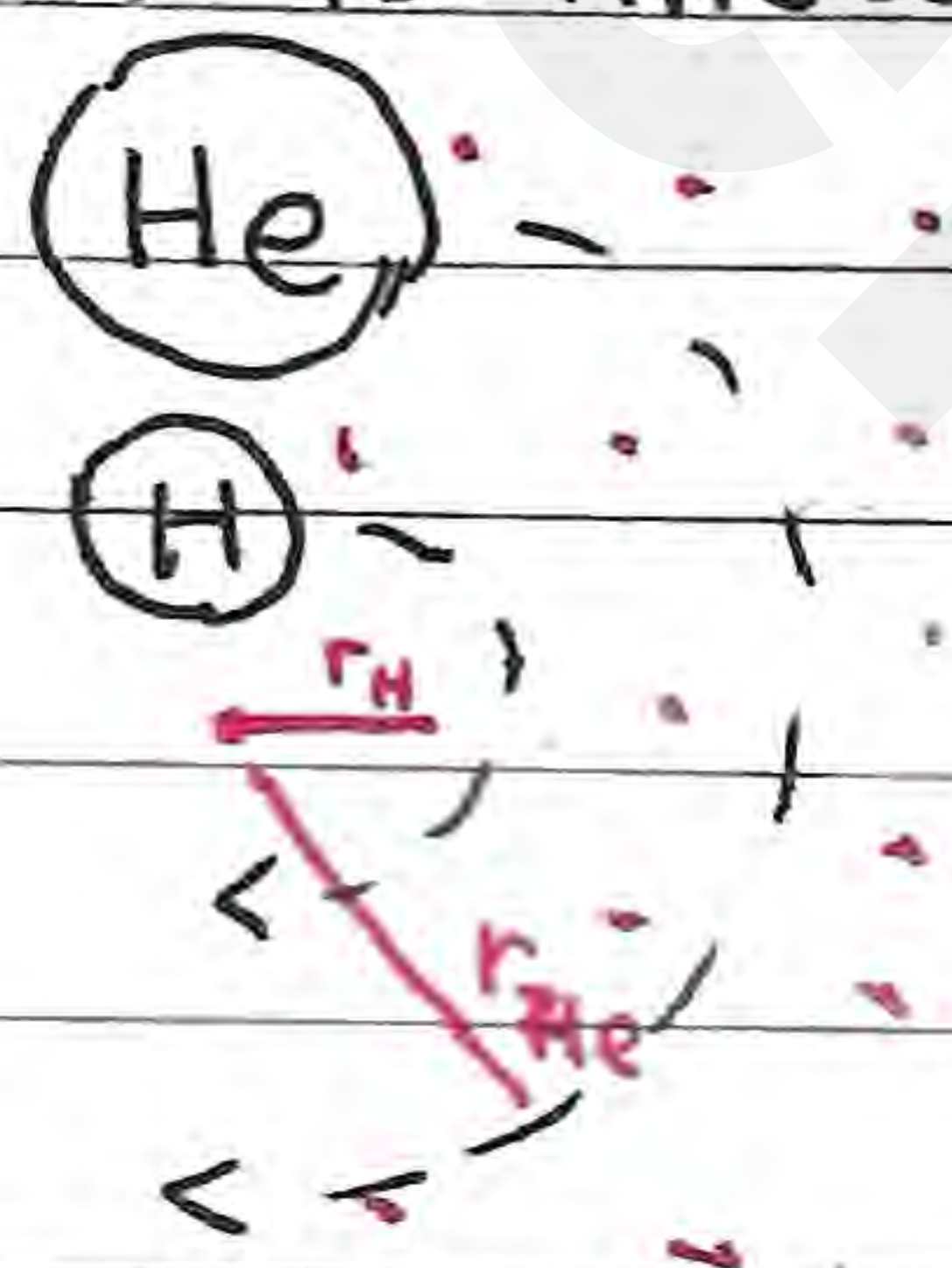
Magnetic Force

When velocity is not perpendicular to magnetic field, we only use the component that is (the parallel component remains unaffected)



Using $r = \frac{mv}{qB}$, we can relate $\frac{m}{q}$ with radius

- This is known as a mass spectrometer



While He has $2 \times q_H$, it also is $4 \times m_H$ (H has no neutrons)
 $\rightarrow$ Thus $r_{He} > r_H$

With this new formula, we can define Amperes more accurately

- 1 Ampere is the current in two 1m wires that are 1m apart with equal currents where the force on them is $2 \cdot 10^{-4} \text{ N}$

Magnetic Force and circular motion

As previously discussed, $r = \frac{mv}{qB}$

We can connect it to kinetic energy

$$E_k = \frac{p^2}{2m}, \quad r = \frac{p}{qB} = \frac{\sqrt{2E_k m}}{qB}$$

Thus $r \propto \sqrt{E_k}$

In this kind of circular motion, kinetic energy stays constant...

• m and v^2 stay constant (mass and ^{not} velocity / speed)

OR - No displacement

No work done

↙ charge over mass

Charge-Mass Ratio (specific charge)

$$\frac{q}{m} = \frac{v}{rB} \quad \text{from previously derived formula}$$

But we can also connect to voltage, for...

$$qV = E_k = \frac{1}{2}mv^2 \quad \leftarrow \text{moving from pt 1 to pt 2}$$

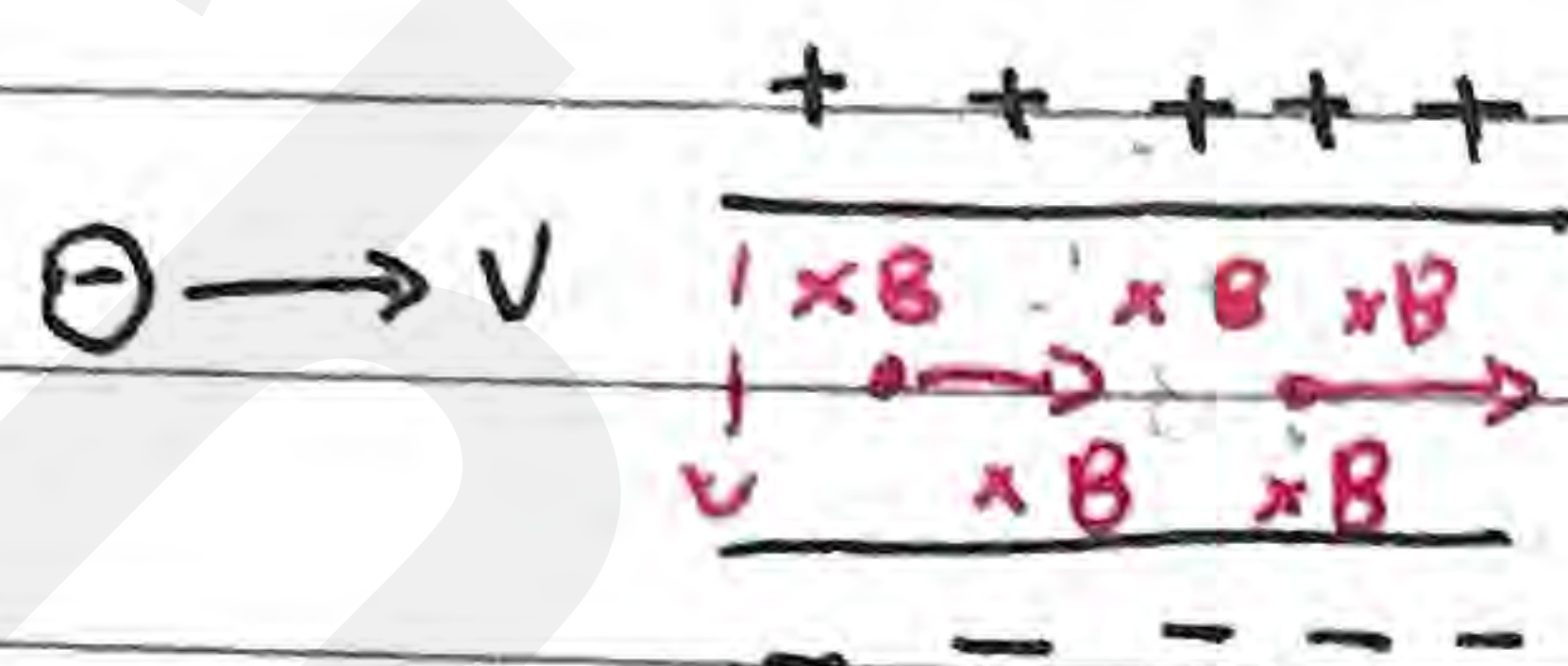
↙ experiment to calculate charge-mass
the slope of the linear line B^2 vs V

$$\frac{q^2}{m^2} = \frac{2qV}{m} \cdot \frac{1}{r^2 B^2} \Rightarrow \frac{q}{m} = \frac{2V}{r^2 B^2}$$

rearranging we could also get $B^2 = \frac{2m}{qr^2} V$

Electric and magnetic fields, moving charge

In a parallel plate with magnetic fields, it is possible to obtain a situation such that the net force is zero (velocity selector)



according to the left-hand rule, B and E cancel out in this scenario

$$F_E = F_M \Rightarrow qE = qvB \Rightarrow v = \frac{E}{B}$$

↳ occurs when E, B adjusted to specific ratio

Why is this useful?

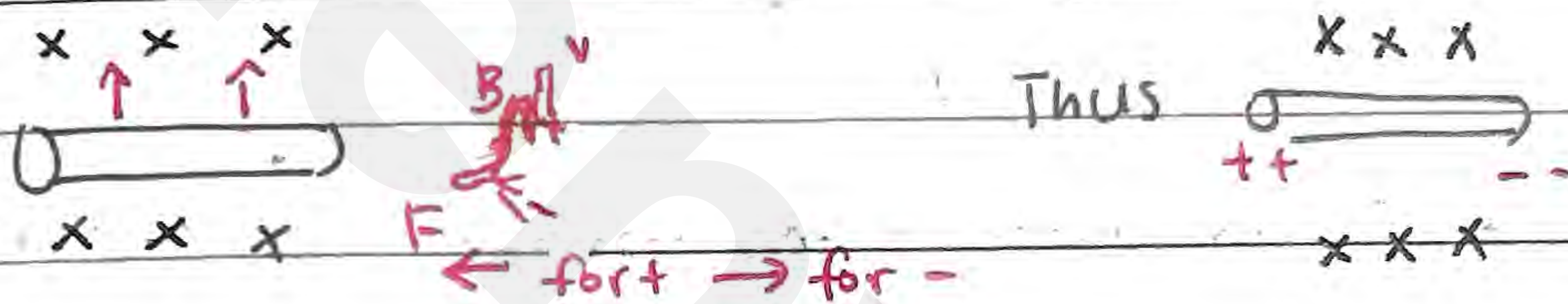
- It allows for selecting particles in one specific direction (since behavior varies, some E greater, some B greater, etc. the ones that do not follow the wanted velocity will collide to the plate)

↳ Ensures only velocities wanted are passed through (direction)

Induced EMF

Moving charges can create current, but can the opposite occur?

Suppose you have a conductive rod in a magnetic field. By moving the rod, the charges move with the rod, causing a force which moves + charges to one end of the rod, - charges to the other.



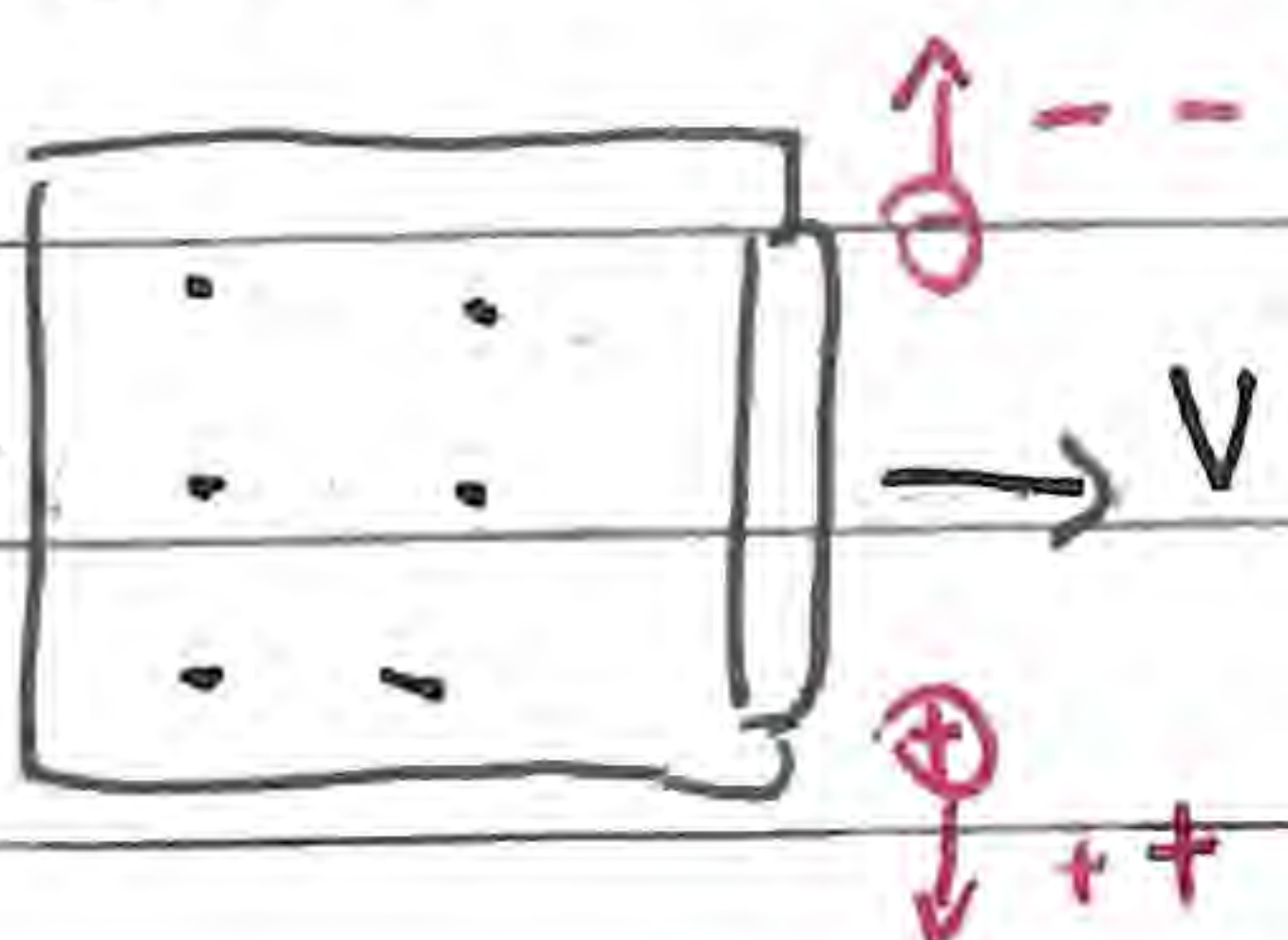
This induces a potential difference, an EMF

Motional emf :
$$\mathcal{E} = v B L$$

Annotations: v is velocity, B is magnetic field strength, L is effective length. $\mathcal{E}$ is EMF.

Analyzing rods in motion

given



Find direction of $\mathcal{E}_m$

To get this, get current by focusing on a single charge

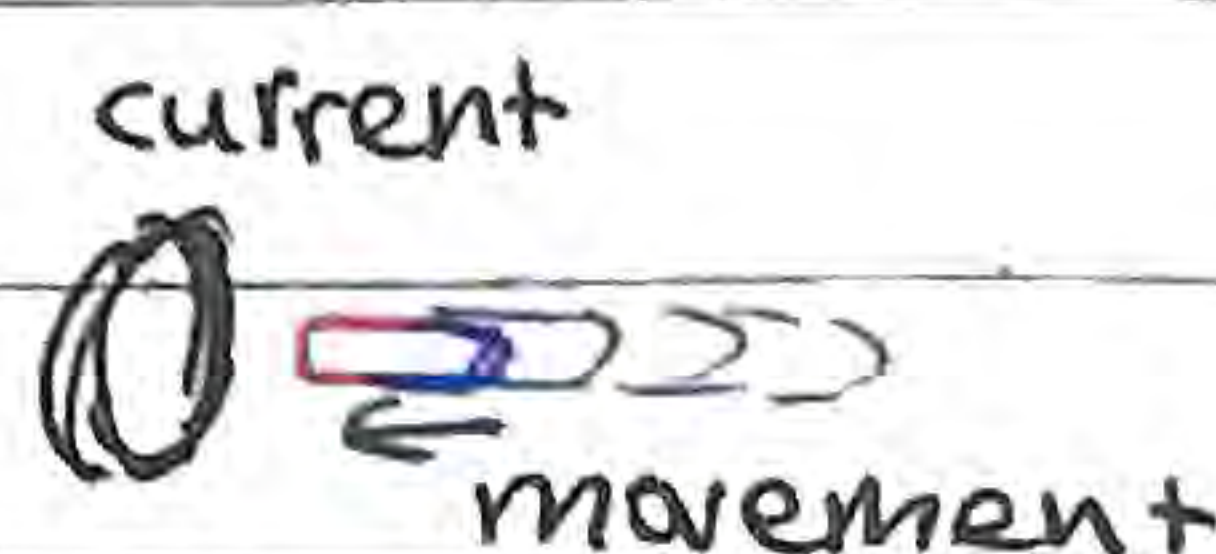
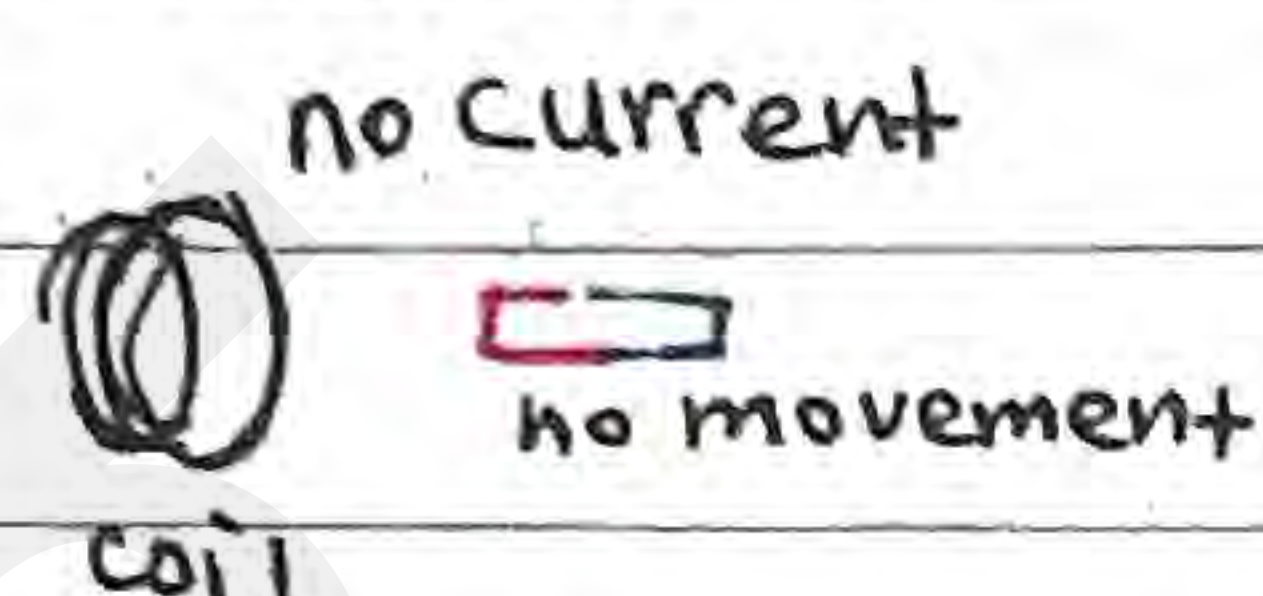
so current $\downarrow$ (use left-hand rule)

$\hookrightarrow$ use left-hand rule again

Faraday's Law

There is an induced EMF when there is a change in magnetic field

• Take a coil...



$$\mathcal{E} = - \frac{\Delta \Phi}{\Delta t}$$

$$\mathcal{E} = - \frac{d\Phi}{dt}$$

← negative of slope

area inside coil

Φ - Magnetic flux, defined $\Phi = B \cdot A \cdot \cos \theta$

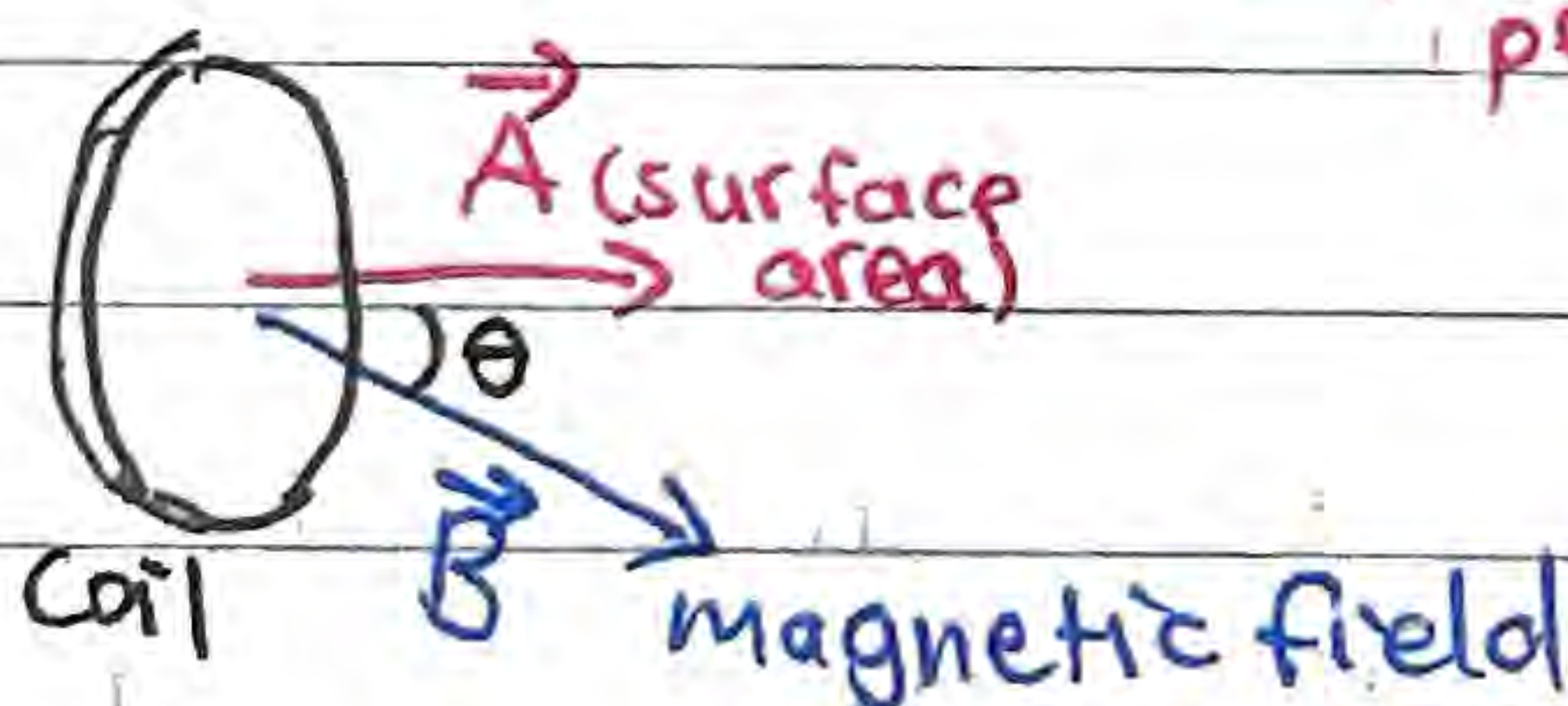
t - Time (Δt is time of movement / changing flux)

angle between $\vec{B}$ and $\vec{A}$

But what is magnetic flux?

• You can think of amount of magnetic field passes through surface area

→ actually vectorial quantity, perpendicular to surface



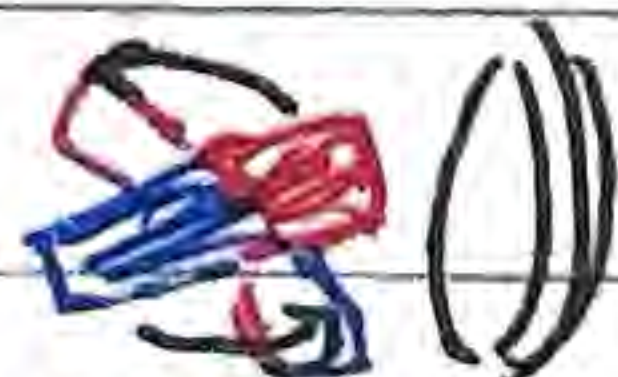
Thus $\Phi = B \cdot A \cdot \cos \theta$

Component of B passing through A .

In real life, B and A are really hard to control

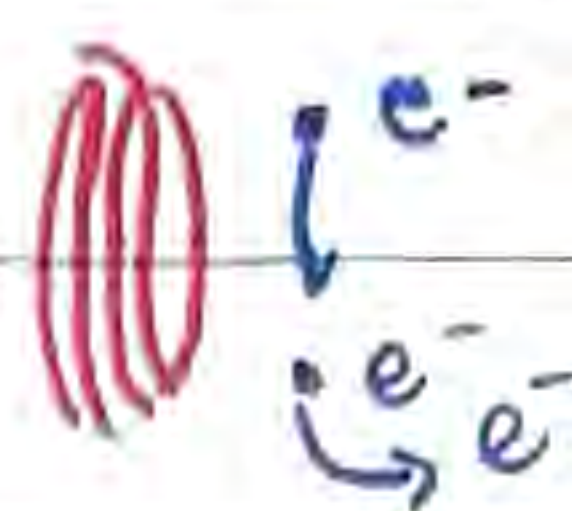
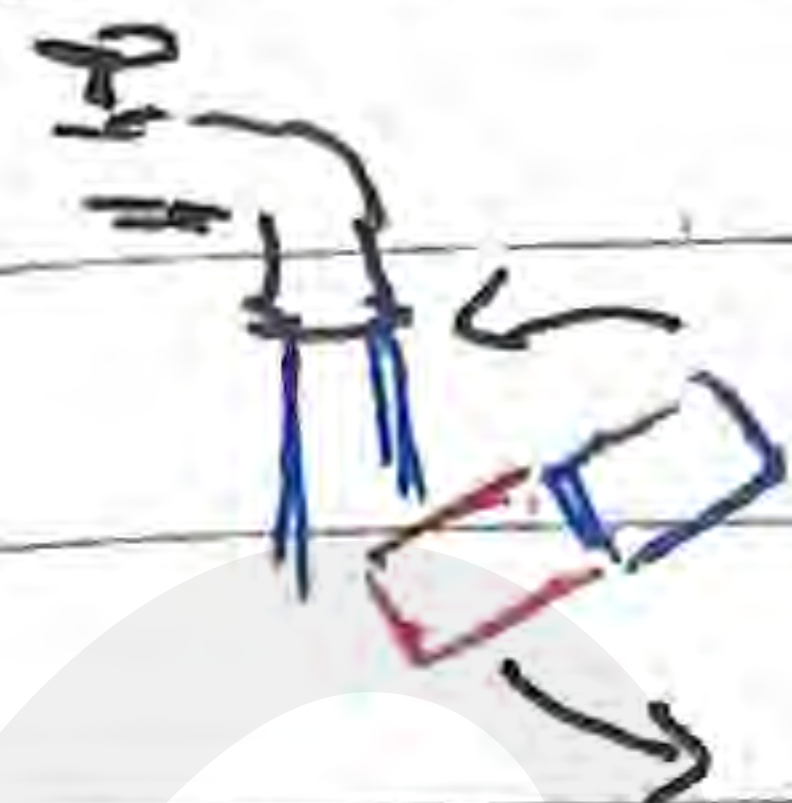
• But θ is easier

↳ with this, current fluctuates between directions (Alternating current)



Faraday's Law

Hydroelectric Power Plant



then



so
current
alternates



Total energy transferred through this calculate $\mathcal{E}$,
then $\text{Power} = \frac{\mathcal{E}^2}{R}$, then $E = P \cdot t$

> Magnetic Flux Linkage $= N \cdot \Phi$

- Considering # of loops in coil (N)
- Thus EMF induced

$$\mathcal{E} = - \frac{\Delta(N \cdot \Phi)}{\Delta t}$$

why "negative"?

Lenz's Law

Application of conservation of energy

- "The induced emf will always act to oppose the change that is producing it."

E.g.

What it really means is $\Delta E = 0$, so any factor affecting

emf must be counteracted

BA ↓
add more x

- Having less field lines in region means $B \cdot A \downarrow$

thus
clockwise

> Thus $B \cdot A$ must be increased somehow, so find which

way you "add" magnetic field, find current with right-hand rule

- Same thing when $B \cdot A \uparrow$

↳ This is induced current

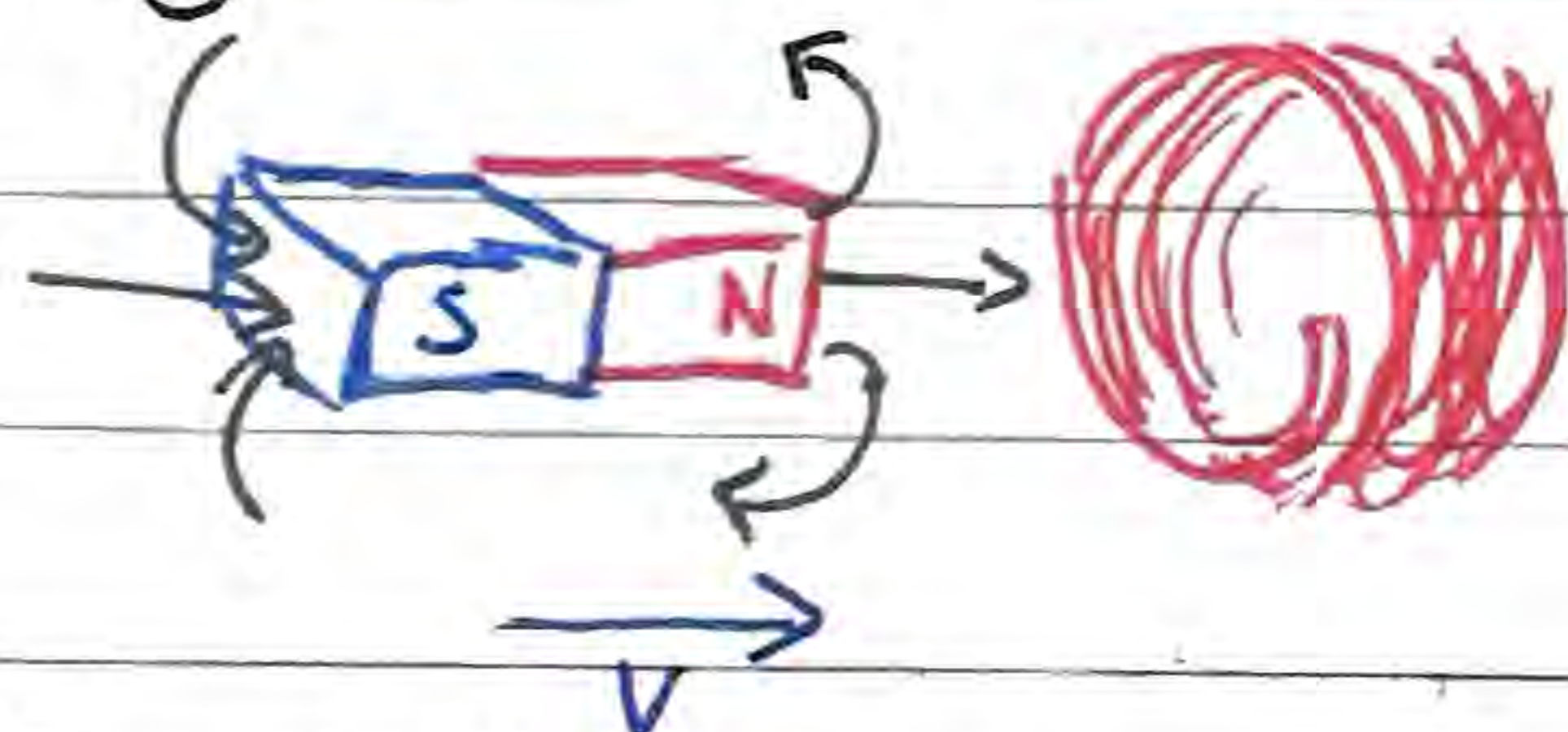
Lenz's Law

- Moving a magnet in a coil induces current and emf
 - > When there's movement, there's current and emf
 - > When there's no movement, there's no current nor emf

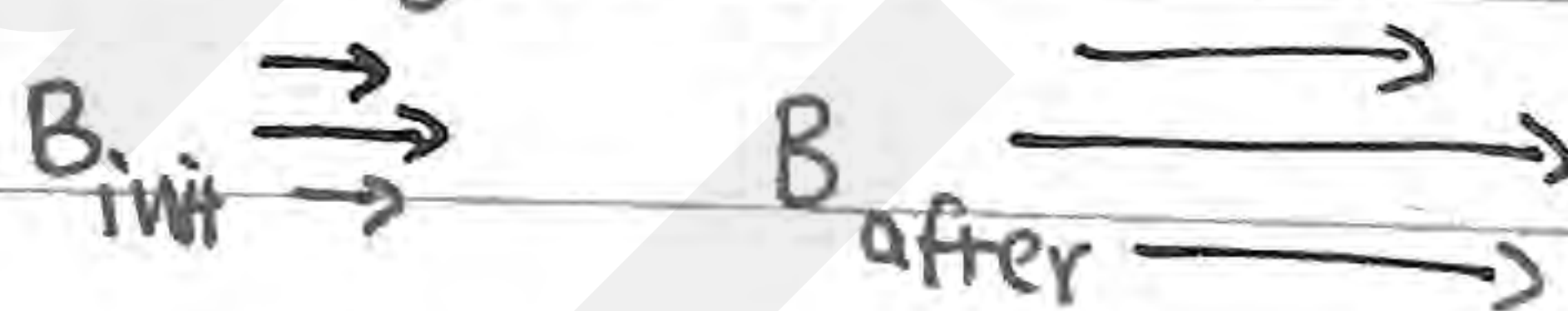
But you must consider conservation of energy...

- When you change magnetic flux, you're "catching" different amount of magnetic field lines $\Rightarrow$ changes B and thus E_{magnet}
- To counteract this, current is induced
 - > Current creates magnetic field lines
 - This means you need to have current to produce magnetic field lines that cancel the change in B

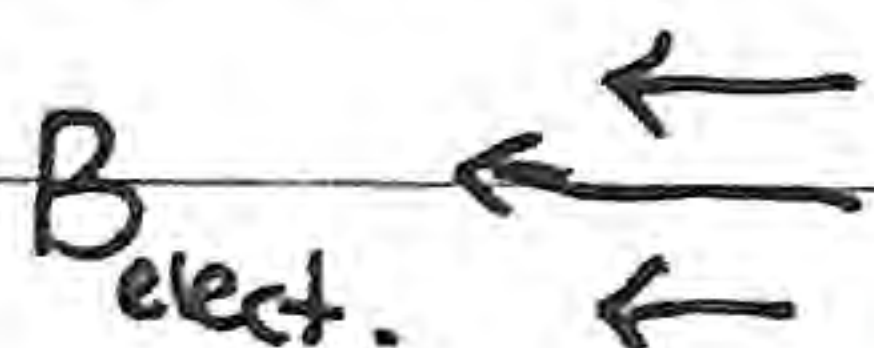
E.g.



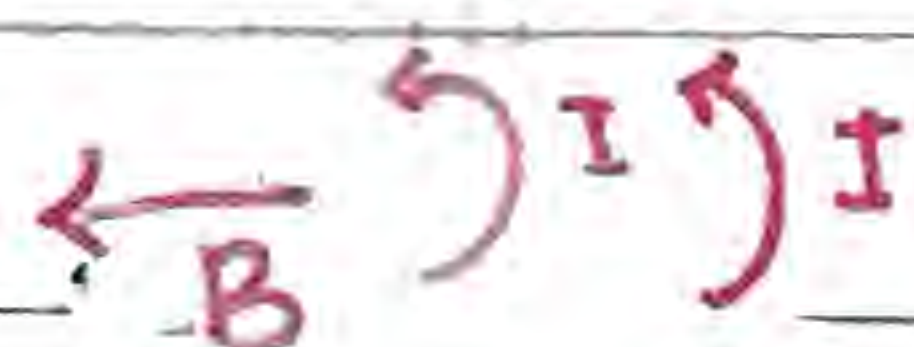
what's change in B ?



So, electric current cancels this so $\Delta B_{\text{net}} = 0$

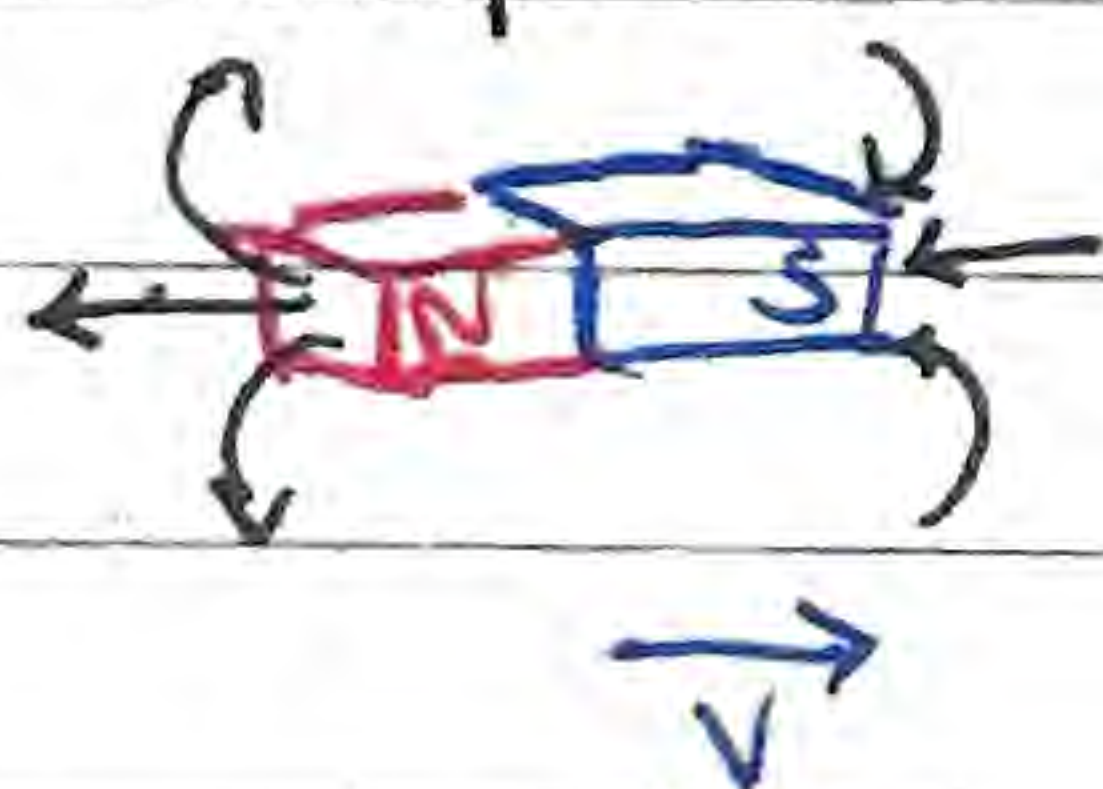


what direction of current produces this?



thus induced current is $\curvearrowright$ (counterclockwise)

Similarly...



ΔB $\leftarrow$ to $\leftarrow$
thus B_{elec} $\rightarrow$



so $\curvearrowleft$ I (clockwise)

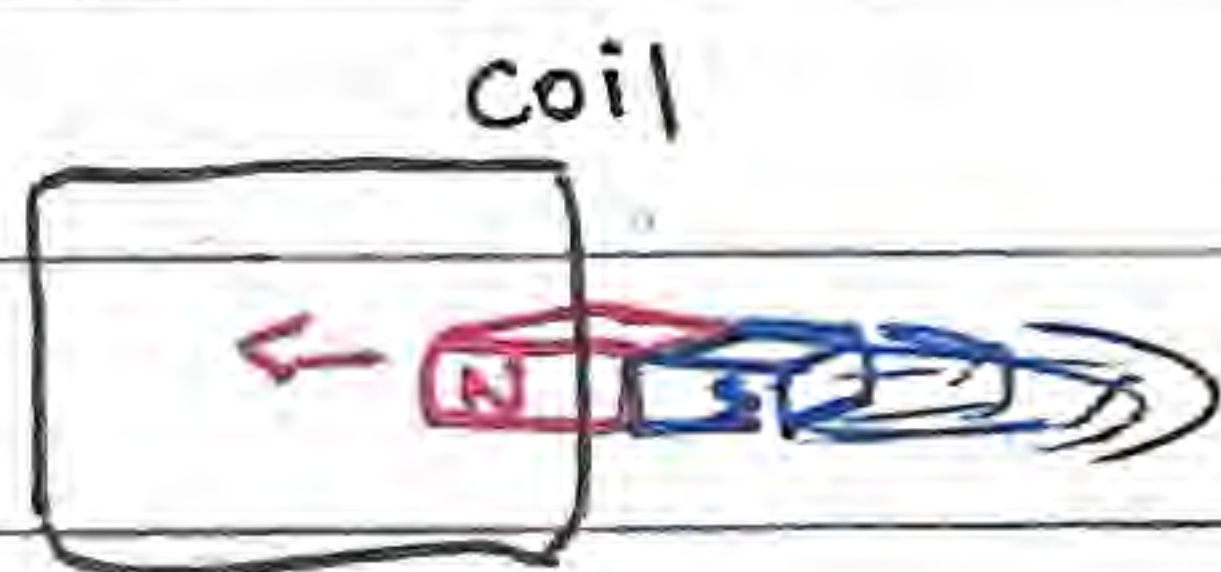
Lenz's Law

The direction of an induced electric current opposes the change in magnetic flux that caused it

- Why conservation of energy?
- The opposition towards magnetic flux change is a resisting motion from induced current
- The extra work you do in pushing the magnet into the coil is energy converted into electrical energy

As such, nature opposes change

E.g.



What's the change?

- Greater field strength

To oppose this change, current is induced anti-clockwise

- opposing magnetic field

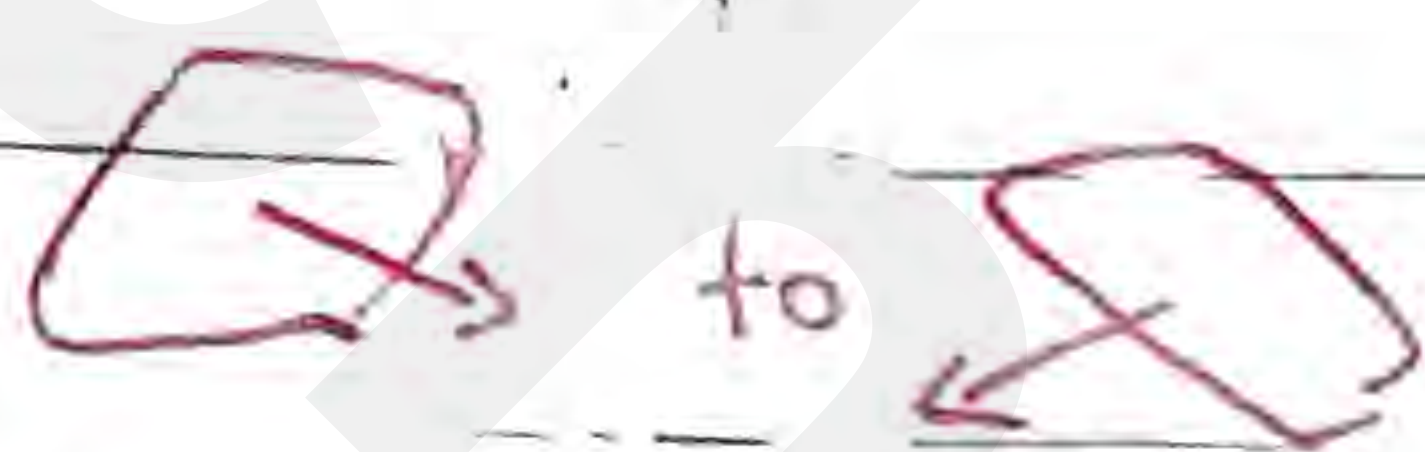
* $\Delta B_{\text{net}} \neq 0$, since resistance prevents ideal induced current

↳ Thus induced magnetic field only partially cancels with change in magnetic field

AC Generators

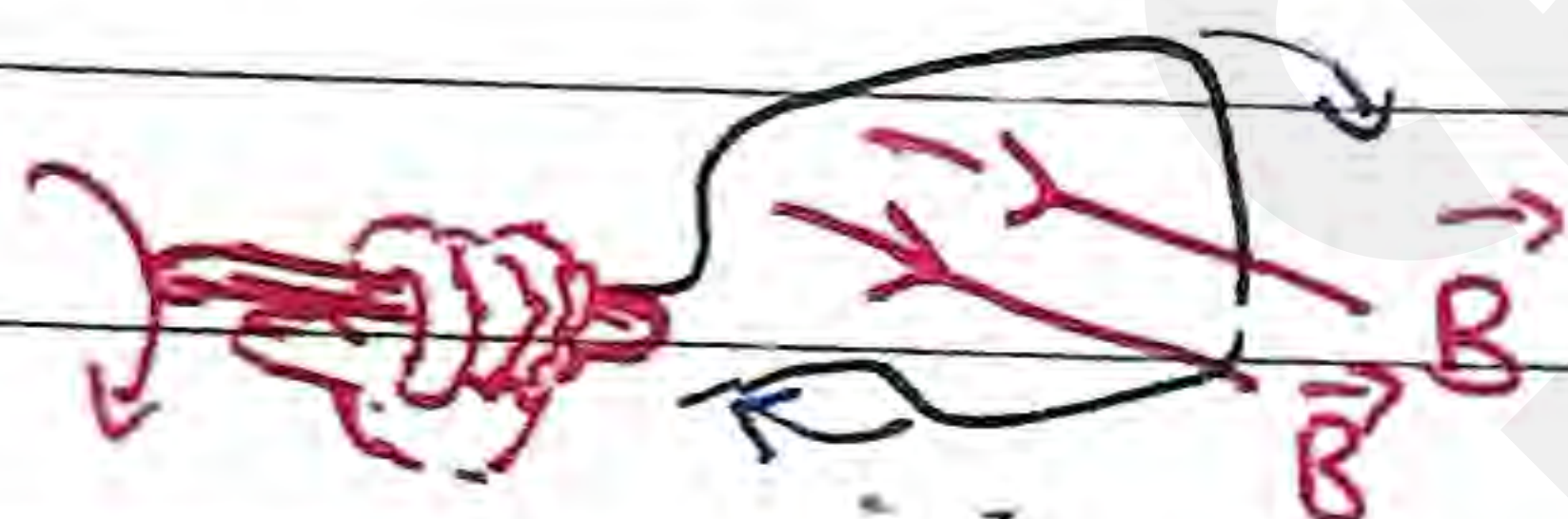
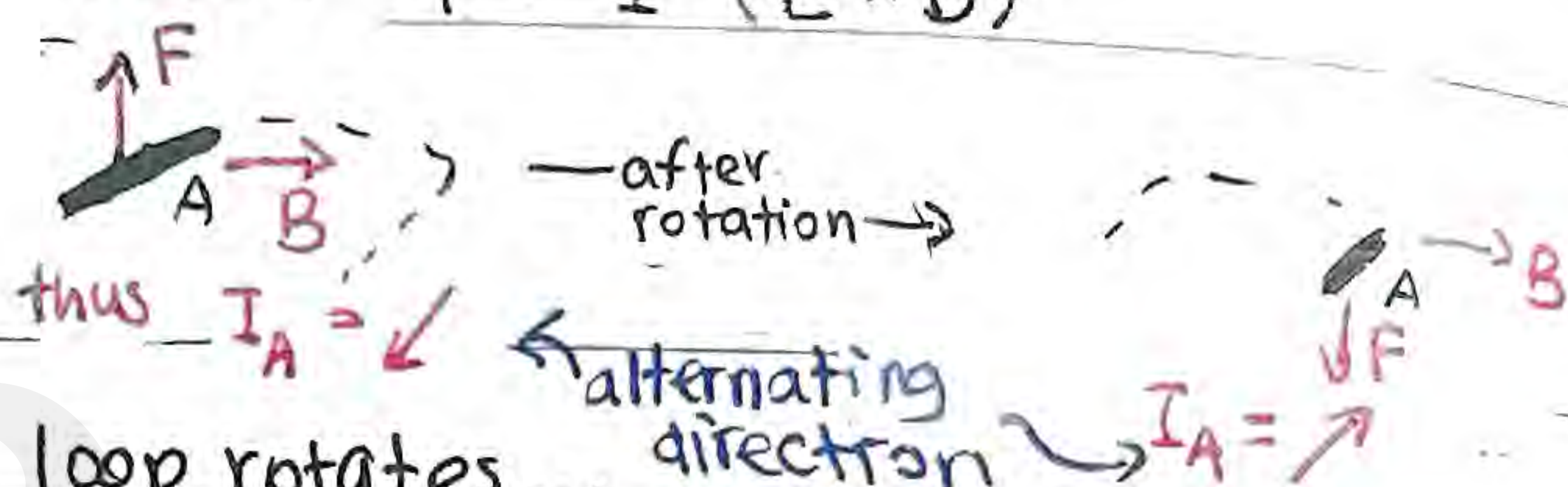
Inputting rotational mechanical energy can generate flux, thus inducing current...

A rotating loop changes $\vec{A}$, changing Φ and thus causing $\mathcal{E} \neq 0$



> Why is it alternating?

Recall: $\vec{F} = I \cdot (\vec{L} \times \vec{B})$

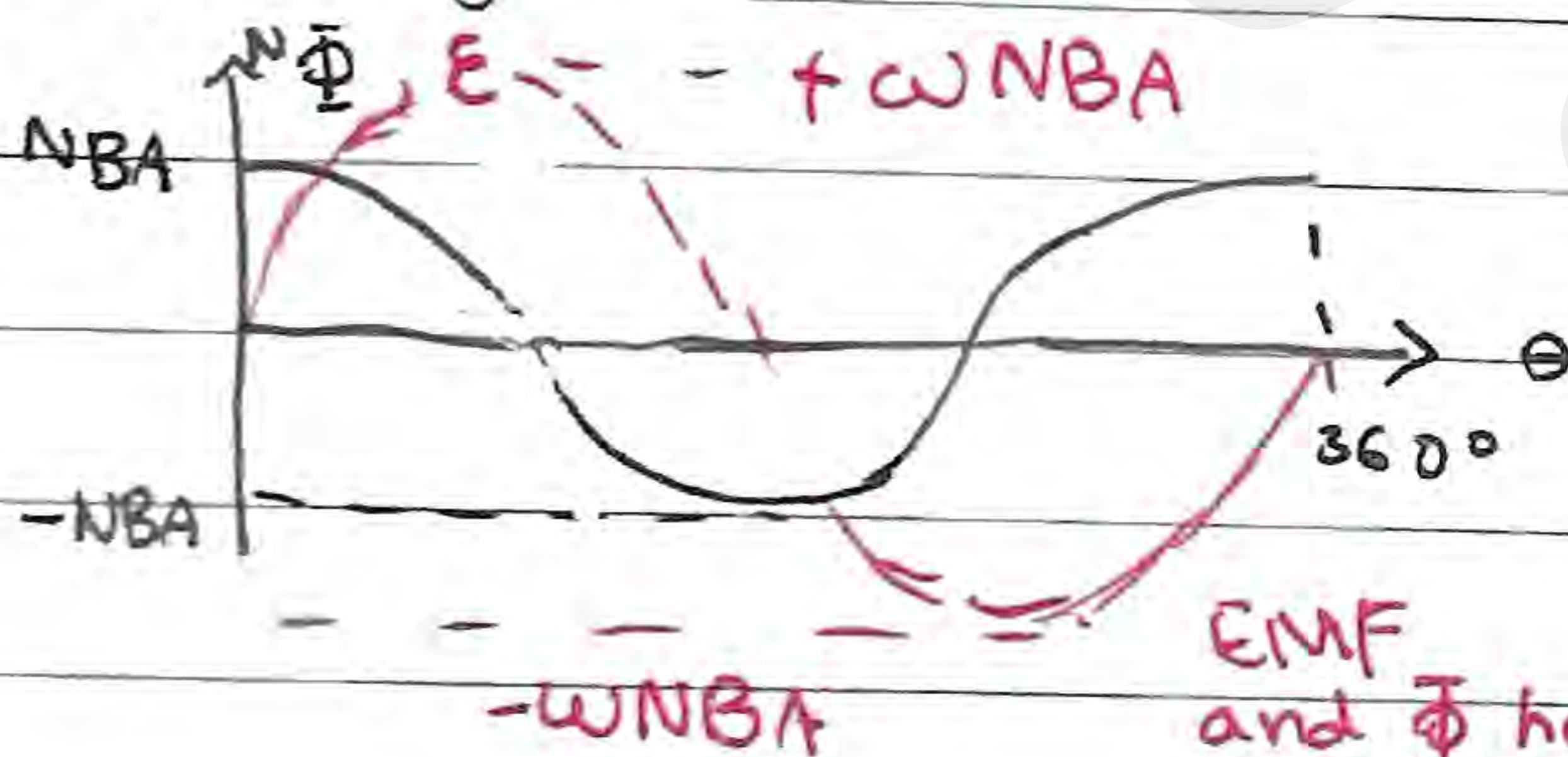


As loop rotates...



$\theta = 0^\circ$ $\theta = 90^\circ$ $\theta = 180^\circ \dots$

forming...



since $\theta = \omega t$

thus

$$\Phi = BA \cos \theta$$

$$= BA \cos(\omega t)$$

EMF and Φ have same frequency/period,

but not necessarily equal amplitudes

in terms of the induced emf...

$$\mathcal{E} = - \frac{d\Phi}{dt} = \omega N B A \sin(\omega t)$$

This differential relation explains how at maximum flux, $\mathcal{E} = 0$

• Since $\mathcal{E}$ fluctuates between < 0 and > 0 , AC current is induced

Since $V = IR$, induced current $I = \mathcal{E} / R$ (I has same freq. as $\mathcal{E}$)

Induced Power

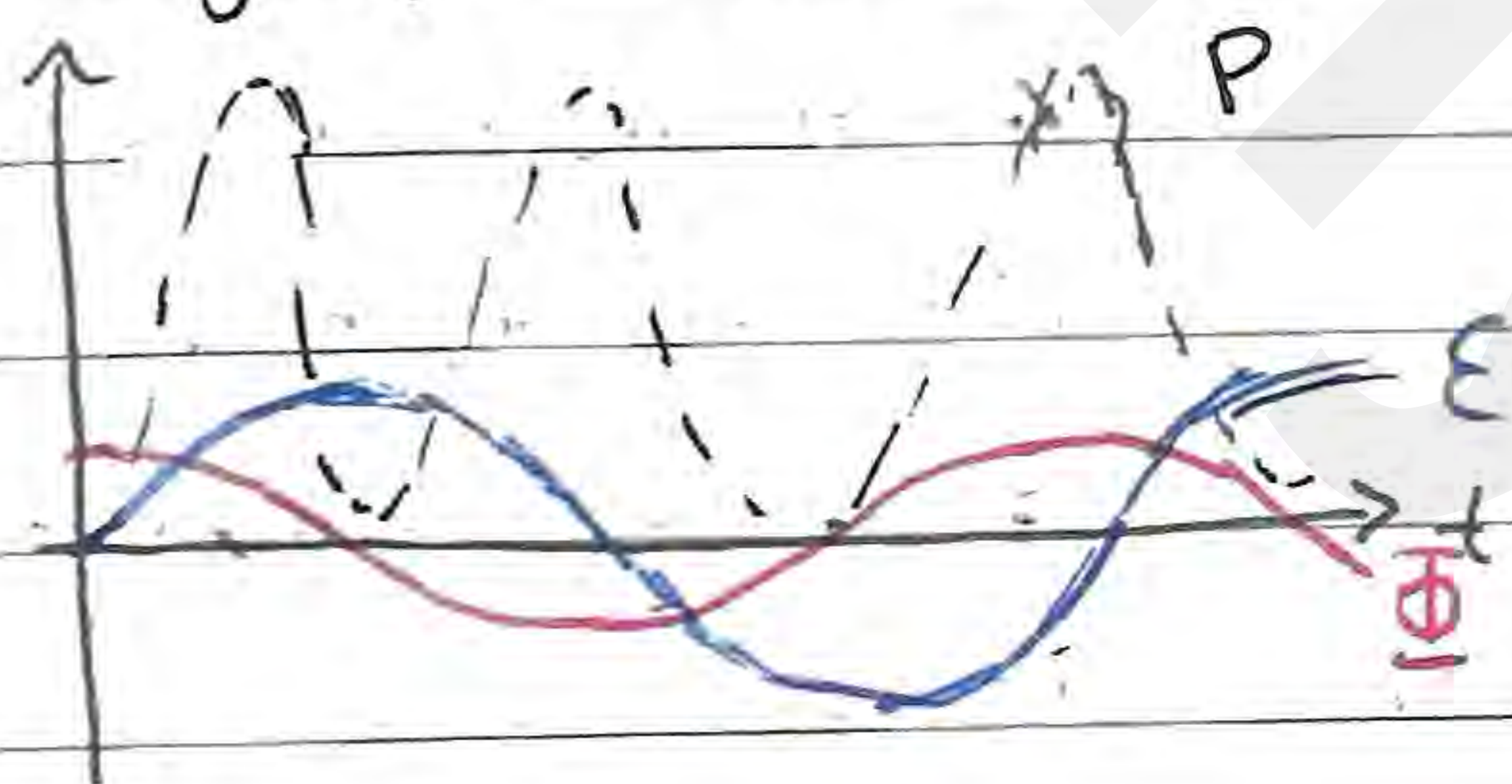
Since $P = \frac{V^2}{R} = \frac{\epsilon^2}{R} = \frac{(-\omega NBA)^2}{R} \sin^2(\omega t)$

- Notice frequency of power $\neq$ frequency of ϵ
- The squared make $P > 0$
- Thus



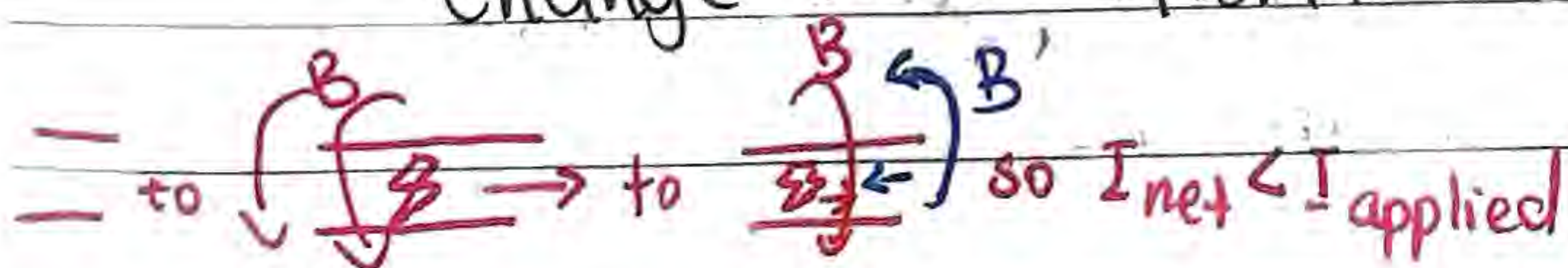
amplitude = $(\omega NBA)^2 / R$
period = period $\epsilon / 2$

The graphs combined form...



Self-induction

- Induced emf can arise in single coils or solenoid
- Lenz's law: $\rightarrow$ current supplied change induces magnetic field
 $\rightarrow$ Thus Δ magnetic flux, Lenz's law induces current
- When current supplied to the coil changes, an opposing change occurs from induced current



- This phenomenon occurs by sockets

(sudden change in flux by unplugging device can induce EMF)

rapid voltage spike